

Multiple Choice

1. **Answer:** A

Options: A) 10; B) 5; C) 0.5; D) 3; E) 2

Note: The capital/output ratio measures the amount of capital used per unit of output produced, and since there are four quarters in a year, an annual capital/output ratio of 2 translates to a quarterly ratio of 2 multiplied by 4, resulting in a ratio of 10.

2. **Answer:** E

Options: A) Smith's principle is outdated; B) Smith's principle relies on the assumption of unlimited natural resources; C) Government intervention is always necessary; D) The "invisible hand" is countered by the presence of monopolies and oligopolies; E) Perfect competition is required for Smith's principle to be completely effective

Note: The correct answer highlights that Smith's principle of the "invisible hand" relies on the existence of perfect competition, which is not present in the U.S. economy today due to factors such as monopolies, market power, and externalities that distort market outcomes.

3. **Answer:** A

Options: A) Real interest rate decreases by 2%; B) Real interest rate more than doubles; C) Real interest rate decreases; D) Real interest rate increases; E) Real interest rate is halved

Note: The real interest rate is calculated by subtracting the inflation rate from the nominal interest rate, so with the nominal rate increasing from 8% to 10% and inflation rising from 3% to 5%, the real interest rate decreases from 5% to 5%, resulting in a net decrease of 2%.

4. **Answer:** A

Options: A) A consumer gets a pizza for free.; B) The federal government has taken in more revenue than it has paid out in expenditures.; C) The total amount a consumer is willing to pay for a pizza is less than the market price.; D) A consumer pays less for a pizza than the cost of its production.; E) It refers to a consumer who no longer has any outstanding debts.

Note: The correct answer exemplifies consumer surplus because consumer surplus is defined as the difference between what a consumer is willing to pay for a good or service and what they actually pay; in this case, receiving a pizza for free indicates a maximum surplus since the consumer pays nothing while valuing the pizza.

5. **Answer:** C

Options: A) unemployment will rise.; B) taxes will have to be raised.; C) the equilibrium price level will rise.; D) government spending on important programs will have to be cut.

Note: The correct answer is based on the fact that expansionary fiscal policy, such as

increased government spending or tax cuts, can lead to higher aggregate demand, which may push the equilibrium price level up as the economy moves toward full employment, potentially resulting in inflationary pressures.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the present value is calculated using the formula $PV = FV / (1 + r)^n$, where FV is the maturity value, r is the yield, and n is the number of years until maturity, and applying this formula with the given values results in approximately \$96,330.28.

7. **Answer:** True

Options: A) True; B) False

Note: An expansionary monetary policy typically lowers interest rates, which can lead to a depreciation of the domestic currency, making exports cheaper for foreign buyers and imports more expensive, but in certain conditions, it can result in a stronger currency, reducing net exports as domestic goods become pricier for foreign markets.

8. **Answer:** True

Options: A) True; B) False

Note: The statement is true because when Bank A receives a 1000 deposit, it is required to keep only a fraction of it as reserves.

9. **Answer:** False

Options: A) True; B) False

Note: A 2-sided alternative test is designed to detect deviations in both directions from the null hypothesis, and it can provide sufficient evidence to reject the null hypothesis if the data shows a significant effect in either direction.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because, in monopolistic competition, firms face low barriers to entry, which encourages competition and allows for product differentiation among similar goods.

Long Form Response

11. **Answer:** Marginal cost (MC) equals average variable cost (AVC) and average total cost (ATC) at their minimum points. This occurs because when MC is less than AVC or ATC, the averages are decreasing, while when MC is greater, the averages are increasing. The significance of these intersection points lies in their role in production analysis; they indicate the most efficient scale of production. At these points, the firm is operating at its optimal output level, where resources are being utilized effectively, and costs are minimized. Thus, understanding these intersections is crucial for firms

in making production decisions and maximizing profitability.

Note: correct because it accurately identifies that marginal cost intersects average variable cost and average total cost at their minimum points, indicating the optimal production level where average costs are minimized, which is crucial for efficient resource allocation in production analysis.

12. **Answer:** The purchase of a used car, such as a 1957 Chevy from a friend, does not contribute to the Gross Domestic Product (GDP) because GDP measures the value of newly produced goods and services within a specific time frame. In this case, the transaction is a transfer of ownership of an existing asset, which does not reflect new production or economic activity. While the sale may involve a price that reflects the car's market value, it does not generate new output or income that would be counted in GDP calculations. Additionally, certain transactions, such as the sale of used goods, are excluded from GDP to avoid double counting, as these goods have already been accounted for in previous production figures. Therefore, while the purchase may have personal economic implications for the buyer and seller, it does not impact the overall GDP.

Note: correct because GDP only accounts for the value of newly produced goods and services, and the sale of a used car represents a transfer of ownership rather than new economic production.